

Motion Orchestration in Dual-Stage Wafer Scanners

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Abstract—In semiconductor manufacturing, lithography machines are becoming more and more sophisticated system of systems. As an example, a TWINSCAN wafer scanner machine is composed of a wafer, and reticle handlers, reticle, optics, and two wafer chains or systems. In previous studies, we covered the interactions between the reticle, optics, and wafer chains during the step-and-scan cycle. In this study, we focus on the interaction between the additional wafer chain responsible for aligning the wafer substrate and taking its height map during the measurement cycle, and the other chains that are active during the step-and-scan cycle. Working in parallel to increase machine throughput, the inertial forces associated motion of the two cycles induce vibration that may propagate throughout the chains in the machine if no appropriate measures are taken. In this investigation, we look at the reference trajectories responsible for steering the chains throughout the two cycles, and propose two reference trajectory orchestrations that factor in the machine design, geometry, mass distribution, and functions. Theoretically, these orchestrations lead to suppressing the induced vibration without sacrificing the machine throughput while keeping the involved control loops intact.

I. INTRODUCTION

Dual-stage wafer scanners combine the precise utilization of motion and light to produce nano-metric integrated circuits (ICs). Compared to other single-stage wafer scanners used in the semiconductor manufacturing, the dual-stage wafer scanners have higher throughput, and therefore, studying their performance is more challenging. Both types of scanners project IC patterns designed on a quartz plate, known as *reticle* [1] onto a silicon disk, known as *wafer*, which is coated with a light sensitive material [2] during the step-and-scan motion. A rectangular area on the wafer substrate known as *die* will emerge as a result of the scanning (exposure) process done under constant velocity. In principle, the reticle is handled by a reticle chain that moves in a direction opposite to the chain handling the wafer substrate. To generate extra dies, the wafer chain moves the wafer substrate to the next die position (stepping process) [3]. This cycle of step-and-scan repeats until the desired locations on a wafer substrate are fully exposed. During rapid production cycles, the machine undergoes high-acceleration levels inducing vibration and performance degradation. To overcome

such degradation, various contributions were reported in the literature [4]–[9].

As depicted in Fig. 1(a), a dual-stage wafer scanner executes multiple tasks in parallel to increase the machine throughput. Namely, step-and-measure, step-and-scan, swap, load-unload wafer substrates and reticle patterns are the main tasks involved in such a machine. During the load-unload tasks, the handler robots do interact with the wafers and reticle stages as needed. During the step-and-measure task, the wafer stage handling the wafer substrate to be aligned and measured interacts directly with the measurement system- a non-motion-related system- where the surface map is used to minimize the leveling errors of the wafer surface topology before the exposure takes place [1]. Similarly, during the step-and-scan task, the reticle, the wafer and the optics chains interact [4], [5], [7]. Sharing the same time and physical space, chain interaction is inevitable during the aforementioned tasks. At all times, the non-motion-related chain, i.e., the optics chain, must be held stationary [10]. To that end, balance masses are usually used with extreme caution about their needed physical workspaces [Chapter 9.2.1] [11].

According to Newton-Euler equations of motion, the motion of a body can be characterized by the driven stage mass properties, applied forces and moments. Besides inducing linear motion, when applied at a distance from the stage center of gravity (CoG), the forces create moment and-based on the available degrees of freedom- stimulate angular motion that is responsible for the angular vibration and the undesired motion-coupling between the interconnected stages. The same can happen at the chains level when they faithfully follow poorly-designed reference trajectories. Usually, the reference trajectories can be designed using various input shaping techniques [12], [13].

To the best of the authors' knowledge, the literature is lacking studies that cover the intertwining of the driven machine geometry, the interconnected stages, and the designed reference trajectories. To that end, in this investigation, we revisit the fillet profile [5] to add a *geometrical aspect* to its definition, and to access its applicability to steer the stages mainly during the parallel execution of the step-and-measure and the step-and-scan tasks in a dual-stage wafer scanner.

In this paper we present a geometry-based input shaping that also factors in the machine design, geometry, and tasks which makes it suitable for multi-stage motion orchestration, multi-tasking, execution and prioritization. The main idea of the proposed orchestration method is to obtain stationary CoG despite of the number of involved motion stages and tasks scheduling. The problem formulation is outlined in

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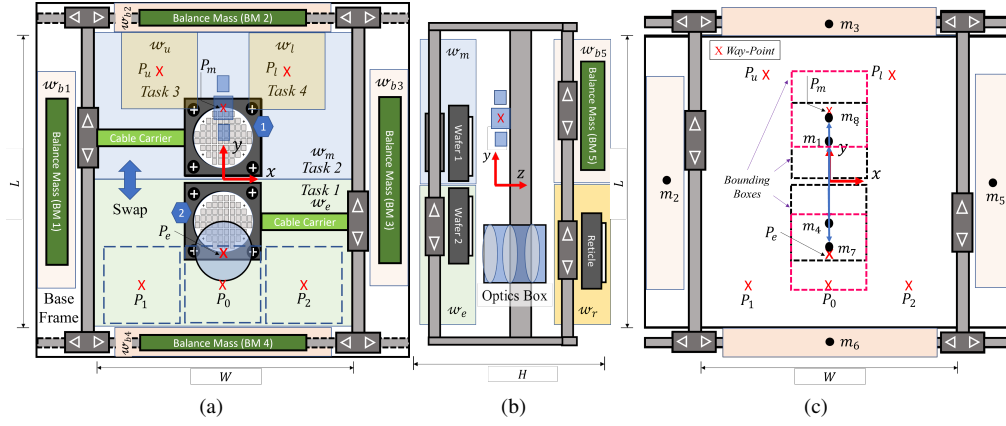


Fig. 1: The (a) top view of the working spaces associated with the involved tasks, i.e., w_e associated with Task 1 during which exposure is done at point P_e , w_m associated with Task 2 during which alignment and height measurements take place at point P_m , w_u associated with Task 3 during which unloading of exposed wafer substrates takes place at point P_u , and w_l associated with Task 4 during which loading of new wafer substrates is done at point P_l . Note that $w_u, w_l \subset w_m$, (b) side view of the machine, (c) way-points and bounding boxes of the motion stages.

Section II, simulation results are provided in Section III. Final remarks are provided in Section IV. Further issues will be handled in a separate study.

II. PROBLEM FORMULATION

A generic lithography machine with N -stages can be characterized by- see [4] for details

$$\begin{aligned} \dot{\mathbb{P}} &= \mathbf{A}_0 \mathbb{P} + \mathbf{A}_1 (\boldsymbol{\Omega}, \dot{\boldsymbol{\Omega}}) \mathbb{P} + \mathbf{B} \mathbf{u} \\ \mathbf{m} &= \mathbf{I} \dot{\boldsymbol{\Omega}} + \mathbf{D} \boldsymbol{\Omega} \\ \mathbf{Y} &= \mathbf{C} \mathbb{P} \end{aligned} \quad (1)$$

where $\mathbb{P} = [\mathbf{P}^T, \dot{\mathbf{P}}^T]^T$ with $\mathbf{P} = [\mathbf{f}_1^T, \mathbf{p}_1^T, \dots, \mathbf{p}_q^T]^T$ comprising the stages positions and velocities, the control input vector $\mathbf{u} = [\mathbf{u}_1^0, \mathbf{u}_1^0, \dots, \mathbf{u}_q^0]^T$, the angular velocity vector $\boldsymbol{\Omega} = [\boldsymbol{\omega}_1^T, \boldsymbol{\Omega}_1^T, \dots, \boldsymbol{\Omega}_q^T]^T$, the inertia matrix $\mathbf{I}$, the inertia cross terms matrix $\mathbf{D}$, the moment vector $\mathbf{m}$, and the sensor matrix $\mathbf{C}$. The system model (1) represents a linear parameter-varying system where the matrix $\mathbf{A}_0$ holds the nominal terms, i.e. when both the angular velocity ($\boldsymbol{\Omega}$), and the angular acceleration ($\dot{\boldsymbol{\Omega}}$) are both zeros, and the matrix $\mathbf{A}_1$ holds the varying parameters, i.e. when $\boldsymbol{\Omega}, \dot{\boldsymbol{\Omega}} \neq 0$. The coupling term given by the matrix $\mathbf{A}_1$ in (1) not only prevents the machine from *ideally* being treated as a linear time-invariant system (LTI)- see [14], but also deteriorates its performance [4]. Ideally, only linear motion exists during machine operation. At the stage level, this can be guaranteed by the adequate distribution of existing actuators. At machine level, the chains movement may induce moments due to inadequately planned reference trajectories that excite the undesired angular motion according to (1).

Consider Fig. 1, the motion of the two wafers chains are constrained to move in only the x - y plane, while the reticle

chain moves only along the y -direction during the scanning motion. Five balance masses (BM) are shown, specifically $BM1$ and $BM2$ are associated with the first wafer stage (Wafer 1) and move along the x -direction and y -direction, respectively. Similarly, $BM3$ and $BM4$ are associated with the second wafer stage (Wafer 2) and move along the x -direction and y -direction, respectively. The reticle $BM5$ is allowed to move only in the y -direction. The whole system can be approximated as a system of particles with point masses- see [14]- located at the center-of-mass of their corresponding stages as depicted in Fig. 1(c), and therefore, Newton's second law is applicable. The net force about the machine CoG is given as

$$\begin{aligned} \mathbf{F}_{CoG}(t) &= \sum_{i=1}^{N=8} \mathbf{F}_i(t) = \sum_{i=1}^{N=8} m_i \ddot{\mathbf{p}}_i(t) \\ \mathbf{M}_{CoG}(t) &= \sum_{i=1}^{N=8} \mathbf{p}_i(t) \times \mathbf{F}_i(t) \end{aligned} \quad (2)$$

where $\mathbf{F}_i$ denotes the force applied to the i^{th} particle whose *desired* position is given by $\mathbf{p}_i = [p_{x_i}, p_{y_i}, p_{z_i}]^T$, and mass is denoted by m_i . Note that we have $\ddot{p}_{z_i} = 0, i = 1, 2, \dots, 8$ denoting wafer 1, BM1, BM2, wafer 2, BM3, BM4, reticle, and BM5 accelerations in the z -direction, respectively. Additionally, we have $p_{x_3} = p_{x_6} = p_{x_7} = p_{x_8} = 0$, $p_{y_2} = p_{y_5} = 0$, and stages 1 through 6 share the same z -value, i.e., in the same plane, while $p_{z_7} = p_{z_8}$. Without loss of generality, let $m_1 = m_4$. Since the reticle moves four times faster than the wafer during scanning and in the opposite direction [8], we have $\ddot{p}_{y_4} = -\alpha \ddot{p}_{y_7}$ where α denotes the demagnification factor used, and which is usually equal to 0.25 [8]. In this study, we have the balance masses actuated and accelerate in the opposite direction relative to their associated stages.

Taking $\ddot{p}_{x_2} = -\frac{m_1}{m_2}\ddot{p}_{x_1}$, $\ddot{p}_{y_3} = -\frac{m_1}{m_3}\ddot{p}_{y_1}$, $\ddot{p}_{x_5} = -\frac{m_4}{m_5}\ddot{p}_{x_4}$, $\ddot{p}_{y_6} = -\frac{m_4}{m_6}\ddot{p}_{y_4}$, $\ddot{p}_{y_8} = -\frac{m_7}{m_8}\ddot{p}_{y_7}$, both $F_{CoG}(t)$ and $M_{CoG}(t)$ in (2) as zeros, and solving for $\ddot{p}_i(t)$, yields

$$F_{CoG}(t) = \mathbf{0}$$

$$M_{CoG}(t) = \mathbf{0} = \begin{bmatrix} 0 \\ 0 \\ \{p_{x_1}\ddot{p}_{y_1} - p_{y_1}\ddot{p}_{x_1}\} + \{p_{x_4}\ddot{p}_{y_4} - p_{y_4}\ddot{p}_{x_4}\} \end{bmatrix} \quad (3)$$

which indicates that under such a generous distribution of balance masses, any designed trajectory can be used as long as it eliminates or at least minimizes the moments about the z -axis. Generally, the i^{th} wafer stage positioning is achieved using a long and short stroke positioning system. Therefore, $p_{x_i} = X_i + x_i$, and $p_{y_i} = Y_i + y_i$ where (X_i, Y_i) is the position due to the long stroke, and (x_i, y_i) is the position due to the short stroke. In this study, we attribute the positioning of the stages to the motion of mainly the long strokes, see [4].

To solve for $M_{CoG}(t) \equiv \mathbf{0}$, $\forall t \geq t_0$ with t_0 denoting the absolute time along the current task time horizon, we may take $p_{x_1} = p_{x_4} \rightarrow p_{y_1} = -p_{y_4}$, or $p_{x_1} = -p_{x_4} \rightarrow p_{y_1} = -p_{y_4}$ and solve the resulting nonlinear second-order ordinary differential equation (ODE). Also, a candidate trajectory that can be used to that end is obtained via solving the differential equations appearing in the first bracket of (3) to handle the step-and-scan task, and by solving the other one in the

second bracket to handle the step-and-measure task, such that p_{x_i} and p_{y_i} are linearly independent for $i = 1, 4$. The ODE given in (3) can be solved using $p_{y_i} = \sin(\omega t)$ and $p_{x_i} = \sin(\omega t + \Theta_i)$, with $i \in \{1, 4\}$, however, the x - y trajectory of this solution will not be *practically* useful when zero initial conditions are used. Using nonzero initial conditions will result in a family of candidate solutions- *with residual moment*- and is given as

$$\begin{aligned} p_{y_i}(t) &= -\gamma_i R_{y_i} \sin(\omega_i t + \Phi_i) + c_{y_i} t + \tilde{c}_{y_i} \\ p_{x_i}(t) &= -\beta_i R_{x_i} \sin(\omega_i (t - \Delta) + \Psi_i) + c_{x_i} t + \tilde{c}_{x_i} \end{aligned} \quad (4)$$

where $\gamma_i, \beta_i \in \{1, -1\}$ determines the direction of stepping and scan/measure movements over the wafer substrate, R_{x_i}, R_{y_i} determines the step size in the x and y -directions respectively, and $\omega_i \leq \min\{\sqrt{a_{x_i}^{max}/R_{x_i}}, \sqrt{a_{y_i}^{max}/R_{y_i}}\}$ with a_{x_i, y_i}^{max} denotes the maximum admissible acceleration in the x, y -directions of the i^{th} wafer stage with $i \in \{1, 4\}$ where $a_{y_i}^{max} \geq a_{x_i}^{max}$ usually holds true. Bounds on the position higher-derivatives can also be introduced. Note that when p_{x_i} and p_{y_i} are linearly dependent, the motion in the x - y plane will be in a straight line, while the tasks require the motion to be linear in the y -direction during the scan/measure periods. Two examples of continuous-time piecewise trajectories are depicted in Fig. 2. Using (4), the stepping motion of the step-and-scan and the step-and-measure tasks using the *symmetric* sin – sin trajectory is given during the nonzero acceleration periods for other than the first die as

$$\begin{aligned} \ddot{p}_{y_i}(t) &= \begin{cases} \gamma_i \omega_i^2 R_{y_i} \sin(\omega_i t) & , 0 \leq t \leq T_i \\ -\gamma_i \omega_i^2 R_{y_i} \sin(\omega_i (t - T_i - D_i)) & , T_i + D_i \leq t \leq 2T_i + D_i \end{cases} \\ p_{y_i}(t) &= \begin{cases} -\gamma_i R_{y_i} \sin(\omega_i t) + \gamma_i R_{y_i} \omega_i t + y_{0_i} & , 0 \leq t \leq T_i \\ \gamma_i R_{y_i} \sin(\omega_i (t - T_i - D_i)) + \gamma_i R_{y_i} \omega_i (t + D_i) + y_{0_i} & , T_i + D_i \leq t \leq 2T_i + D_i \end{cases} \\ \ddot{p}_{x_i}(t) &= \begin{cases} -\beta_i \omega_i^2 R_{x_i} \sin(\omega_i t) & , 0 \leq t \leq T_i \\ \beta_i \omega_i^2 R_{x_i} \sin(\omega_i (t - T_i - D_i)) & , T_i + D_i \leq t \leq 2T_i + D_i \end{cases} \\ p_{x_i}(t) &= \begin{cases} \beta_i R_{x_i} \sin(\omega_i t) + \beta_i R_{x_i} \omega_i t + x_{0_i} & , 0 \leq t \leq T_i \\ -\beta_i R_{x_i} \sin(\omega_i (t - T_i - D_i)) + \beta_i R_{x_i} \omega_i (t + D_i) + x_{0_i} & , T_i + D_i \leq t \leq 2T_i + D_i \end{cases} \end{aligned} \quad (5)$$

where $\omega_i = \pi/T_i$, and (x_{0_i}, y_{0_i}) holds the initial position of the current task and the position of the i^{th} wafer stage CoG (C_{x_i}, C_{y_i}) with respect to the machine CoG. Substituting (5) into (3) yields the residual moment of the i^{th} wafer stage in the z -direction (M_{z_i}) given as

$$M_{z_i}(t) = \begin{cases} \omega_i^2 m_i(t) \sin(\omega_i t) & , 0 \leq t \leq T_i \\ -\omega_i^2 m_i(t) \sin(\omega_i (t - T_i - D_i)) & , T_i + D_i \leq t \leq 2T_i + D_i \\ 0 & , \text{otherwise} \end{cases} \quad (6)$$

where $m_i(t) = 2\gamma_i \beta_i R_{x_i} R_{y_i} \omega_i t + \gamma_i R_{y_i} x_{0_i} + \beta_i R_{x_i} y_{0_i}$ with $i \in \{1, 4\}$. Therefore, the overall residual moment at the machine level is given as $M_{CoG_z}(t) = M_{z_1}(t) + M_{z_4}(t)$,

which suggests that both tasks, i.e. the step-and-scan and step-and-measure should extend over the same time horizon in case we desire $M_{CoG_z}(t) = 0$ - if possible- under the sin – sin designed trajectories. At the die level, taking $T = \max\{T_1, T_4\}$, and set the measure time D_1 equal to the exposure time $D_4 \equiv D$ during which the die is subjected to high energy that needs to be minimized [5], results in $\omega_1 = \omega_4 \equiv \omega$. Using (6), we have

$$M_{CoG_z}(t) = \begin{cases} \omega^2 m(t) \sin(\omega t) & , 0 \leq t \leq T \\ -\omega^2 m(t) \sin(\omega (t - T - D)) & , T + D \leq t \leq 2T + D \\ 0 & , \text{otherwise} \end{cases} \quad (7)$$

where $m(t) = m_1(t) + m_4(t)$.

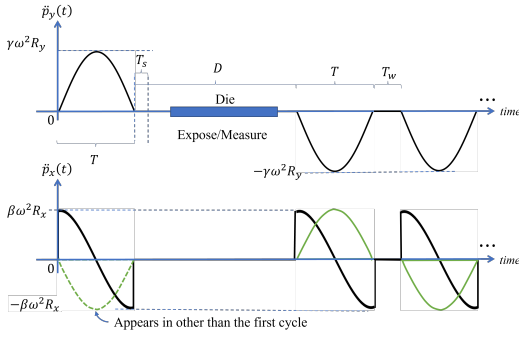


Fig. 2: Two examples of the trajectories given by (4) $\ddot{p}_y - \ddot{p}_x$, i.e. $\sin - \sin$ and $\sin - \cos$, where $R_y = R_x = \gamma = 1$, the cycle $\forall t \in [0, 2T + D + T_w]$, and $\omega = \pi/T$.

A. Step, Scan, and Measure Pitch Sizes

During both the step-and-scan and step-and-measure, a stepping phase is involved which allows moving between dies across the wafer substrate either to expose or to map the surface topology. During wafer map measurement, an entire array of dies is measured and then a step motion takes place to the neighboring array, i.e., so-called long scans that take place with fixed routing in 6 strokes to cover the wafer. It was shown that measure profiles do cross-over to exposure and result in performance-relevant error signals [1]. Mathematically, the step, the scan, and the measure pitch sizes of the i^{th} wafer stage under the *symmetric* $\sin - \sin$ trajectory are given as

$$\begin{aligned} S_s P_{step_i} &= 2 R_{x_i} \omega_i (T_i + T_{w_i}) \\ S_m P_{scan/measure_i} &= 2 R_{y_i} \omega_i (T_i + D_i) \end{aligned} \quad (8)$$

where $S_s, S_m > 0$ are scalar weights to adjust the step and the scan- when used to realize stepping in the y -direction- pitch sizes according to the orchestration in place. The *scan/measure speed* ($V_{scan/measure}$) in the y -direction is given as

$$V_{scan/measure_i} = 2 R_{y_i} \omega_i \quad (9)$$

and the light source is controlled accordingly.

B. Zero Residual Moment

Setting $m(t)$ in (7) to zero yields $M_{CoG}(t) \equiv \mathbf{0}, \forall t \geq t_0$. Therefore, a stationary CoG is obtained. To that end, the following must hold

$$\begin{aligned} 2\omega \{ \gamma_1 \beta_1 R_{x_1} R_{y_1} + \gamma_4 \beta_4 R_{x_4} R_{y_4} \} t + \\ \{ \gamma_1 R_{y_1} x_{0_1} + \beta_1 R_{x_1} y_{0_1} + \gamma_4 R_{y_4} x_{0_4} + \beta_4 R_{x_4} y_{0_4} \} &= 0 \end{aligned} \quad (10)$$

Taking $R_{x_1} = R_{x_4}, R_{y_1} = R_{y_4}, \gamma_1 \beta_1 = -\gamma_4 \beta_4, P_{scan} = P_{measure}$, (10) can be rewritten as

$$R_{y_1} \{ \gamma_1 C_{x_1} + \gamma_4 C_{x_4} \} + R_{x_1} \{ \beta_1 C_{y_1} + \beta_4 C_{y_4} \} = 0 \quad (11)$$

which can be solved by taking $\gamma_1 = -\gamma_4, \beta_1 = \beta_4, C_{x_1} = C_{x_4}$, and $C_{y_1} = -C_{y_4}$. Now, the problem melts down to finding $(C_{x_1}, C_{y_1}) \in w_m$ and $(C_{x_4}, C_{y_4}) \in w_e$ such that the two wafer stages, the reticle and BM5 do not collide throughout the active task- see Fig. 1(b) and Fig. 1(c).

C. New Conceptual Design

To stretch the benefits of (2) towards the machine design process, let us consider the case where additional masses are to be totally avoided. The main idea behind the proposed conceptual design depicted in Fig. 3 is to utilize the involved stages as balance masses to each other under a suitably designed motion orchestration that maintains stationary CoG of the machine. Of course, additional steps including feasibility analysis must follow to complete the design cycle, which are beyond the scope of this study. Following the same previous notation when no balance masses are involved, equating (2) to zero yields

$$\begin{aligned} F_{CoG}(t) &= \begin{bmatrix} m_1 \ddot{p}_{x_1} + m_4 \ddot{p}_{x_4} \\ m_1 \ddot{p}_{y_1} + m_4 \ddot{p}_{y_4} + m_7 \ddot{p}_{y_7} \\ 0 \end{bmatrix} = \mathbf{0} \\ M_{CoG}(t) &= \mathbf{0} = \begin{bmatrix} (-m_1 p_{z_1}) \ddot{p}_{y_1} + (-m_4 p_{z_4}) \ddot{p}_{y_4} + (-m_7 p_{z_7}) \ddot{p}_{y_7} \\ (m_1 p_{z_1}) \ddot{p}_{x_1} + (m_4 p_{z_4}) \ddot{p}_{x_4} \\ (-m_1 p_{y_1}) \ddot{p}_{x_1} + (m_1 p_{x_1}) \ddot{p}_{y_1} + \dots \\ (-m_4 p_{y_4}) \ddot{p}_{x_4} + (m_4 p_{x_4}) \ddot{p}_{y_4} \end{bmatrix} \end{aligned} \quad (12)$$

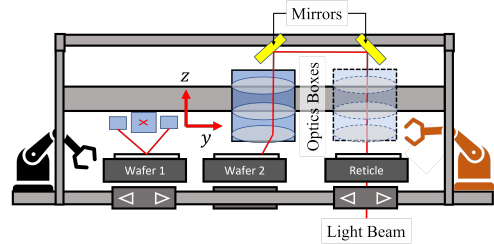


Fig. 3: The proposed conceptual design of a dual-stage wafer scanner with no additional balance masses.

Using $m_1 = m_4, p_{z_1} = p_{z_4}$, and $\ddot{p}_{y_4} = -\alpha \ddot{p}_{y_7}$, and by solving $F_{CoG}(t) = \mathbf{0}$ in (12), we get

$$\ddot{p}_{x_1} = -\ddot{p}_{x_4}, \ddot{p}_{y_1} = \left\{ -1 + \frac{m_7}{\alpha m_1} \right\} \ddot{p}_{y_4} = c \ddot{p}_{y_4} \quad (13)$$

Now, using (13) to solve $M_{CoG}(t) = \mathbf{0}$ in (12), yields

$$\begin{aligned} \ddot{p}_{y_1} &= \left\{ -1 + \frac{m_7 p_{z_7}}{\alpha m_1 p_{z_1}} \right\} \ddot{p}_{y_4} \\ \left\{ p_{y_1} - p_{y_4} \right\} \ddot{p}_{x_4} + \left\{ c p_{x_1} + p_{x_4} \right\} \ddot{p}_{y_4} &= 0 \end{aligned} \quad (14)$$

Equating $\ddot{p}_{y_1}$ in (13) and (14), we find that $p_{z_7} \equiv p_{z_1}$ as depicted in Fig. 3. Moreover, integrating (13) and $\ddot{p}_{y_4} = -\alpha \ddot{p}_{y_7}$ twice yields

$$\begin{aligned} p_{x_1} &= -p_{x_4} + p_1 t + p_0, \quad p_{y_1} = c p_{y_4} + v_1 t + v_0 \\ p_{y_4} &= -\alpha p_{y_7} + q_1 t + q_0 \end{aligned} \quad (15)$$

where p_0, p_1, v_0, v_1, q_0 , and q_1 are constants. Consequently, using (15), the last equation in (14) can be rewritten as

$$\frac{\ddot{p}_{y_4}}{\{c-1\} p_{y_4} + v_1 t + v_0} + \frac{\ddot{p}_{x_4}}{\{1-c\} p_{x_4} + c p_1 t + c p_0} = 0 \quad (16)$$

Let $Y = \{c-1\} p_{y_4} + v_1 t + v_0$, and $X = \{1-c\} p_{x_4} + c p_1 t + c p_0$, therefore, $\forall c > 1 \in \mathbb{R}$ (16) can be given as

$$\frac{\ddot{Y}}{\{c-1\} Y} - \frac{\ddot{X}}{\{c-1\} X} = 0 \quad (17)$$

which has $Y = \gamma \sin(\sqrt{c-1} \omega t)$, and $X = \beta \cos(\sqrt{c-1} \omega t)$ as a solution with $\gamma, \beta \in \mathbb{R}$ are chosen similar to (4), and $\omega = \pi/(T\sqrt{c-1})$. Therefore, we have $\forall t \in [0, T]$

$$\begin{aligned} p_{x_1} &= \frac{\beta}{c-1} \cos(\sqrt{c-1} \omega t) - \frac{p_{11}}{c-1} t - \frac{p_{01}}{c-1} \\ p_{y_1} &= \frac{\gamma c}{c-1} \sin(\sqrt{c-1} \omega t) - \frac{v_{11}}{c-1} t - \frac{v_{01}}{c-1} \\ p_{x_4} &= \frac{-\beta}{c-1} \cos(\sqrt{c-1} \omega t) + \frac{c p_{14}}{c-1} t + \frac{c p_{04}}{c-1} \\ p_{y_4} &= \frac{\gamma}{c-1} \sin(\sqrt{c-1} \omega t) - \frac{v_{14}}{c-1} t - \frac{v_{04}}{c-1} \\ p_{y_7} &= \frac{-\gamma}{\alpha(c-1)} \sin(\sqrt{c-1} \omega t) - \frac{v_{17}}{c-1} t - \frac{v_{07}}{c-1} \end{aligned} \quad (18)$$

Note that $c > 1$ implies that $m_7 > 2\alpha m_1$, with α denoting the demagnification factor of the optics box. Considering (15), the integration constants appearing in (18), i.e. $p_{ij}, v_{ij}, q_{ij}, i \in \{0, 1\}, j \in \{1, 4\}$, can be chosen freely such that

$$\begin{aligned} p_0 &= \frac{c p_{04} - p_{01}}{c-1}, \quad p_1 = \frac{c p_{14} - p_{11}}{c-1} \\ v_0 &= \frac{c v_{04} - v_{01}}{c(c-1)}, \quad v_1 = \frac{c v_{14} - v_{11}}{c(c-1)} \\ q_0 &= -\frac{(v_{04} + \alpha v_{07})}{c-1}, \quad q_1 = -\frac{(v_{14} + \alpha v_{17})}{c-1} \end{aligned} \quad (19)$$

to facilitate designing the the sin-cos profile depicted in Fig. 2. Designing the measurement task profiles in (18) $\forall t \in [0, T]$ results in $p_{01} = \beta, p_{11} = 0, v_{01} = 0, v_{11} = \gamma c \omega \sqrt{c-1}$ with the task parameters given as

$$P_{scan} = \frac{2\gamma c \omega}{\sqrt{c-1}} (T+D), \quad V_{scan} = \frac{2\gamma c \omega}{\sqrt{c-1}}, \quad P_{step} = \frac{2\beta}{c-1} \quad (20)$$

while designing the scan task profiles in (18) $\forall t \in [0, T]$ results in $p_{04} = \beta/c, p_{14} = 0, v_{04} = 0, v_{14} = \gamma \omega \sqrt{c-1}$ with the task parameters given as

$$\begin{aligned} \tilde{P}_{scan} &= \frac{2\gamma \omega}{\sqrt{c-1}} (T+D) = \frac{2\pi \gamma (T+D)}{T(c-1)} \\ \tilde{V}_{scan} &= \frac{2\gamma \omega}{\sqrt{c-1}} = \frac{2\gamma \pi}{T(c-1)}, \quad \tilde{P}_{step} = \frac{2\beta}{c-1} \end{aligned} \quad (21)$$

From (20) and (21), it is clear that the two tasks share the same step size, while the measure task can cover more dies compared to the scan task proportional to the value of $c > 1$. Interestingly, with large c values, the measure task can be finished before the scan task concludes. Having m_1 possibly idle while the measurements are being processed, *switching* to another orchestra involving only m_4 , and m_7 -designed as outlined previously- might be needed.

III. SIMULATION

Taking $C_{x_1} = C_{x_4} = 0.4$ m, and $C_{y_1} = -C_{y_4} = 0.4$ m, and using the *symmetric sin-sin* trajectory (5) with $T = 0.03125$ s, $D = 0.218$ s, $T_w = 0.017$ s, $P_{step} = 0.0262$ m, and $P_{scan} = 0.0623125$ m results in an operation time of 0.2975 s/Die. Equivalently, under the orchestra depicted in Fig. 4, a 300 mm wafer substrate with 60 dies of size equal to 32 mm $\times$ 26 mm has an operation time of 17.85 s/Wafer when the nonproductive stepping phases are excluded. Fig. 5 shows the acceleration profile during the step-and-measure of a complete wafer, where the accelerations duration and maximum values are equal throughout the task to reduce the errors associated with set-point variation [15]. Notably, the reported operation time in Fig. 5 is ≈ 19 s/Wafer or in terms of wafer-per-hour (WPH)- 189 WPH. Therefore, a loss of ≈ 12.7 WPH is expected, and can be minimized by changing the orchestration used, which include trajectory optimization [5]-, and/or allowing higher acceleration levels during stepping compared to scan/measure motions, see Fig. 5- such that the residual vibration is reduced or completely eliminated.

Let $m_1 = m_4 = m_7 = 20$ Kg, every balance mass has a mass of 500 Kg, and $\alpha = 0.25$ be used. Essentially, a BM has to stay within range of its sensors, and it only needs to absorb the stage acceleration forces. Therefore, a BMs homing task should be involved before a new cycle is initiated. Similarly, the motion of the reticle stage during the tasks of interest is given as $p_{y_7} = -\alpha^{-1} p_{y_4}$, which requires that the machine length (L) depicted in Fig. 1(a) is large enough to accommodate the motion of both the reticle stage and BM5 such that $p_{y_7} \notin w_{b5}$ and $p_{y_8} \notin w_r$. This shows the importance of considering the machine geometry when designing the desired trajectories [5]. Fig. 6 shows the scan task profile under the proposed conceptual design

using the previous parameters where (21) is used to find the values of $\gamma = 0.0112$, and $\beta = 0.1179$ for $c = 10$. Having multiple orchestrations capable of achieving the same goal inspires the quest for the *best* one that jointly maximizes the throughput and performance when optimization of the used motion orchestration is carried out under any given machine design, geometry, and operating conditions.

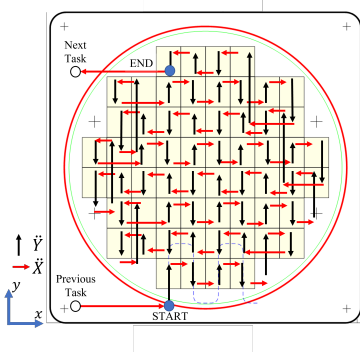


Fig. 4: The proposed step and scan/measure orchestra during the step-and-scan and step-and-measure tasks.

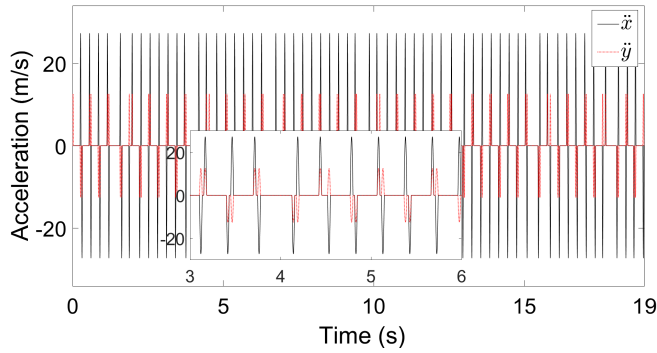


Fig. 5: The x - y accelerations of wafer 1.

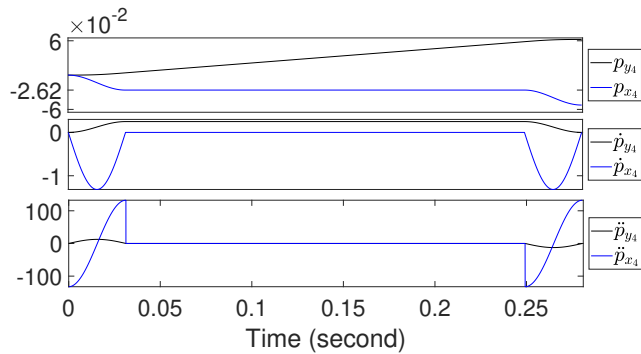


Fig. 6: The resulting scan profile (in meter units) under the orchestra associated with the proposed conceptual design.

IV. CONCLUSION

In this investigation, we approached the reference motion design problem for dual-stage wafer scanners from a geometrical perspective, where the motion stages are treated

ideally as a system of moving point masses. Mainly, two tasks were of interest, step-and-scan and step-and-measure. According to this perspective, a well-designed reference trajectory is capable of steering the motion of the machine while minimizing or completely eliminating any residual vibration, respecting the machine geometrical, structural, kinematical and dynamical features, and meeting other imposed operational constraints. Consequently, upper bounds on the machine throughput and performance can be obtained. When the machine flexible modes are present, any used input shaping technique should yield well-designed reference trajectories. Overall, failing to meet the criteria presented herein, the machine dynamics will misbehave whenever it follows the designed trajectory faithfully, which leads to performance degradation.

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